

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



INTERNSHIP REPORT ON

“The Customer Blueprint: Exploring and Modeling Retail Behavior across Channels”

Submitted in partial fulfillment for the award of the degree (21INT82)

Bachelor of Engineering

In

Artificial Intelligence and Machine Learning

Submitted by:

Charan Gowda B M - 1DS21AI012

Undertaken at



Saks OFF 5TH, Manyata Embassy Business Park, SEZ, G3 Teak,
5th & 6th Floor, Outer Ring Road Bengaluru, India, 560045

**Department of Artificial Intelligence and Machine Learning,
Dayananda Sagar College of Engineering, Bengaluru -**

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CERTIFICATE

This is to certify that the Internship titled **“The Customer Blueprint: Exploring and Modeling Retail Behavior across Channels”** carried out by **Charan Gowda B M [1DS21AI012]**, a bonafide student of **Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the
Reporting Manager
Subhro Mukherjee,
Manager, Data Science,
Saks OFF5th

Signature of the
Internal Guide
Prof. Deeshree B,
Assistant Professor,
Dept. of AI & ML, DSCE

Signature of the
Head of the Department
Dr. Vindhya P Malagi,
Professor & Head,
Dept. of AIML, DSCE

External Viva:

Name of the Examiners

1. _____

2. _____

Signature with date

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



DECLARATION

I, **Charan Gowda B M**, a final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bengaluru, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Intern, Marketing Analytics** at **Saks OFF 5TH, Bengaluru**.

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

Date: 16/04/2025

USN: 1DS21AI012

Place: Bengaluru

NAME: Charan Gowda B M



December 19, 2024
Charan Gowda B M



Internship Letter

We are pleased to offer an Internship to you at Hudson's Bay Services Private Limited, ("HBC" or "the Company") situated at 4th 5th & 6th Floors, Wing A, Block Teak (G3), Manyata Embassy Business Parks Special Economic Zone, Outer Ring Road, Bangalore - 560045 with Technology Team from January 7, 2025 till December 31, 2025.

This letter sets out the main terms and conditions of your internship with the Company-

- **Confidentiality:** During your internship and (if your internship is discontinued for any reason whatsoever) thereafter, you may have access to trade secrets and confidential business information belonging to the Company. By accepting this internship, you acknowledge that you will keep all of this information strictly confidential. Please sign the Non-Disclosure Agreement ("NDA") which is attached as Annexure-1 to this letter. This NDA will form a part of the Internship Letter.
- **Representations:** You represent that: (1) you are not a party to any agreement that would prohibit you from entering into internship with the Company; (2) no trade secret or proprietary information belonging to your previous employers/supervisors will be disclosed by you at the Company and that no such information, whether in the form of documents, memoranda, software, drawings, etc., will be retained by you or brought with you to the Company; (3) you will be required to adhere to various policies and procedures of the Company during the term of your internship; (4) you agree to conduct yourself in a professional manner and in a manner that does not conflict with HBC's interests.
- **Termination of Internship:** You may terminate your internship with HBC at any time upon providing 7 days' advance notice in writing of your last intended day of work. The Company may terminate your internship with or without cause at any time and for any reason. Accordingly, this letter is not to be construed or interpreted as containing any guarantee of continued or permanent employment.
- **Company Property:** You agree that, upon conclusion of your internship, you will immediately return to the Company all of its property, equipment, and documents, including electronically stored information
- You will be entitled to a stipend of **INR 35000** per month from January 2025 to June 2025 and **INR 50000** per month from July 2025 to December 2025, less statutory deductions and other required withholdings as applicable. This will not include any of the employee benefits that regular Company employees receive, including, but not limited to, health insurance, vacation or leave. Please note that your potential conversion to a full-time role as a level 1-Analyst will be subject to an assessment of your performance at the end of your internship period.



- You understand and agree that any breach by you of the provisions in this letter could cause the Company to suffer irreparable harm and no adequate remedy at law would be available in respect thereof. Accordingly, you agree that upon any such breach, the Company shall be entitled to seek relief, as may be granted by a court of competent jurisdiction.

We look forward to having you join us and we expect that our relationship will be mutually rewarding. To confirm your acceptance of this internship, please return a signed copy of this document. to the Company by (validity period/date) confirming your acceptance of the position. If this letter is not received by said date, the internship will be considered retracted.

Best Regards,

A handwritten signature in black ink, appearing to read 'Madhu', followed by a long, sweeping horizontal line that extends to the right.

Madhu Natesan
Vice President, Saks.com, HBC India

Hudson's Bay Services Private Limited

Registered Address: 4th, 5th and 6th Floors of Block Teak (G3), Manyata Embassy Business Park, Special Economic Zone, Nagawara & Rachenahalli Village, Outer Ring Road, Bangalore 560045, Karnataka, India Tel: +91 80 30606000

CIN: U72200KA2016FTC097220

As a Service Provider of Saks

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Aswin S**, Senior Data Scientist, Saks OFF 5TH, for the valuable industry insights and mentorship provided during my role as an intern in Marketing Analytics.

I would also like to extend my thanks to **Prof. Deepshree Buchade**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible

CHARAN GOWDA B M [1DS21AI012]

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CHAPTER 1

INTRODUCTION

The fashion retail industry operates in a dynamic environment where customer preferences shift rapidly across digital and physical platforms. Omni channel strategies have become essential for maintaining customer engagement and delivering a seamless shopping experience. Data analytics plays a vital role in uncovering patterns, preferences, and behaviors in this complex landscape. Retailers increasingly rely on advanced data techniques to retain customers and personalize marketing efforts. Internship projects focused on leveraging analytics to drive customer-centric strategies within this retail framework.

The internship was conducted in the Marketing Analytics department of a fashion retail company with a strong Omni channel presence. The scope of work included data retrieval, cleaning, preparation, and analysis of customer-related datasets. Tasks were aligned with business goals such as improving customer retention, understanding engagement metrics, and modeling future behavior. Practical exposure to real-time data enabled insights into how analytics supports decision-making in retail. The environment provided opportunities to explore both transactional and behavioral data for deeper analysis.

Retention metrics were calculated for the fiscal year to evaluate customer engagement and loyalty trends. Key indicators such as cohort sizes, email subscriptions, app downloads, and loyalty program enrollments were analyzed using historical data. The objective was to assess customer stickiness and identify growth opportunities through channel-based insights. Findings supported strategic planning for marketing interventions and engagement campaigns. The metrics also contributed to identifying effective acquisition and retention channels.

Exploratory data analysis was performed on transactional data to examine the likelihood of customers making a repeat purchase within a given period. Analysis revealed that the majority of repeat purchases occur within 90 days of the previous transaction. This insight led to segmenting customers based on purchase recency and channel preference—online, in-store, or omnichannel. Additional patterns were identified related to purchasing frequency and product preferences across channels. The results enabled the development of timelines for proactive customer engagement strategies.

CHAPTER 2

ABOUT THE COMPANY



Saks OFF 5TH is a leading off-price luxury retailer in North America, widely recognized for delivering premium fashion at significantly reduced prices. It operates as a subsidiary of Hudson's Bay Company (HBC), one of the oldest and most prestigious retail groups globally, with a rich history dating back to the 17th century. Saks OFF 5TH was originally launched as an accessible extension of Saks Fifth Avenue, a luxury department store known for its upscale merchandise and curated fashion. The brand was designed to make designer labels and luxury goods more attainable to a broader audience, without compromising on quality or style.

The company's product portfolio is diverse, offering a wide range of men's and women's apparel, footwear, handbags, beauty products, and home goods. It prides itself on maintaining a strong lineup of internationally acclaimed designer brands, making high fashion more inclusive. This unique model positions Saks OFF 5TH as a powerful player in the retail space, bridging the gap between aspirational luxury and value. Customers are drawn not only by the prestige of the labels but also by the promise of discovering premium pieces at affordable price points, often with new inventory rotating frequently to maintain freshness and urgency.

Saks OFF 5TH has built a robust presence throughout the United States and Canada, operating through a network of physical stores as well as a rapidly growing e-commerce platform. Its Omni channel model ensures that customers experience consistent service across both digital and physical touchpoints. With capabilities such as real-time inventory visibility, click-and-collect, personalized product recommendations, and streamlined return processes, the company focuses on offering a frictionless and satisfying shopping journey. The integration of technology in retail operations not only improves customer experience but also enhances the efficiency of inventory management and order fulfillment.

What sets Saks OFF 5TH apart is its strategic use of data and analytics to drive business decisions. The company leverages advanced analytical tools to gain actionable insights from customer behavior, purchase history, channel performance, and market trends. These insights are used to refine marketing strategies, optimize pricing and promotions, and curate inventory

assortments that align with customer preferences. Personalized marketing campaigns, loyalty initiatives, and retargeting strategies are tailored based on data-driven customer segmentation, which helps in increasing engagement and retention.

Furthermore, Saks OFF 5TH is committed to digital innovation and continuous transformation. As part of a larger push within the retail industry to modernize and personalize the shopping experience, the company invests in enhancing its technology infrastructure. From mobile-optimized shopping platforms to predictive algorithms for customer recommendations, these advancements contribute to a more intelligent and adaptive business model. In an era where consumer behavior is rapidly evolving, Saks OFF 5TH's ability to remain agile through data, innovation, and customer focus is a testament to its strong positioning in the retail landscape.

In addition to its core retail operations, Saks OFF 5TH emphasizes building long-term customer relationships through loyalty programs and personalized engagement. By rewarding repeat purchases and offering exclusive deals to members, the company fosters a sense of community and brand loyalty among its shoppers. These programs not only enhance customer satisfaction but also generate valuable data on shopping habits, enabling the brand to fine-tune its offerings. This focus on customer-centric strategies highlights Saks OFF 5TH's commitment to delivering not just products, but meaningful and tailored experiences. Such initiatives reinforce the brand's competitive edge in the evolving luxury retail market.

CHAPTER 3

SCOPE OF THE WORK

1. Customer Retention Analysis

Customer retention analysis was a fundamental part of the project, aimed at understanding long-term engagement and loyalty patterns within the existing customer base. Various key performance indicators (KPIs) were analyzed, including Average Order Value (AOV), Units per Transaction (UPT), purchase frequency, and engagement levels of email subscribers. These metrics were tracked over time to evaluate the effectiveness of marketing efforts and identify opportunities for optimization in customer retention campaigns.

The analysis provided insights into behavioral differences among customer segments, highlighting which groups demonstrated consistent repeat purchases and which were prone to disengagement. By identifying the characteristics of high-retention customers, the business was better equipped to design personalized engagement strategies and loyalty initiatives. This component of the project laid the groundwork for more targeted interventions and data-backed marketing decisions.

2. Cohort-Based Customer Segmentation

Cohort analysis was employed to segment customers based on their acquisition period and to study their behavior over time. Each cohort represented customers who made their first purchase within the same time frame (e.g., by month or quarter). This approach enabled the identification of retention trends specific to each cohort and helped track how customer behavior evolved across different stages of the customer lifecycle.

The comparison of cohorts revealed fluctuations in retention rates and spending behavior, allowing the business to pinpoint which acquisition campaigns yielded the most loyal customers. Further analysis helped in understanding external factors (such as seasonality or promotional campaigns) that influenced cohort performance. These insights proved valuable for planning future acquisition strategies and tailoring engagement methods for specific customer groups.

3. Churn Prediction Modeling

Churn prediction was another critical aspect of the project, focusing on building a machine learning model to identify customers at risk of leaving the platform. Using historical transactional data and behavioral indicators such as purchase frequency, recency, and average order size, a dataset was prepared for modeling.

Classification algorithms were then used to assign a churn probability to each customer. This predictive model aimed to support the marketing team in proactively identifying at-risk customers before they disengage. By targeting these customers with retention strategies such as re-engagement emails or exclusive offers, the business could reduce churn and improve overall customer lifetime value. The model also provided interpretability by highlighting the most significant factors contributing to churn.

4. Marketing Insights and Business Recommendations

The final phase of the project focused on translating analytical findings into actionable business insights. Reports and visualizations were created to communicate key patterns and trends in customer behavior. These included dashboards highlighting retention metrics, cohort comparisons, and churn risk scores. The outputs were intended to assist marketing and sales teams in designing data-informed strategies for customer retention and acquisition.

Based on the analyses conducted, several recommendations were proposed to improve marketing performance. These included personalized email campaigns for high-risk customers, adjusting promotional timing based on cohort trends, and reallocating marketing budgets toward high-retention segments. The project not only contributed to tactical decisions but also emphasized the importance of continuous data monitoring in building a customer-centric marketing approach.

5. Exploratory Data Analysis of Store Performance

EDA was conducted on store-level data to uncover patterns in customer visits, transaction volumes, and channel preferences (in-store vs. online vs. omnichannel). The aim was to determine behavioral trends that influence purchase decisions across channels. Special attention was given to time between purchases, which helped define the “N” period of 90 days for likely repurchase.

Visualizations and statistical summaries were used to identify peak shopping periods, store performance variance, and customer movement across channels. These insights assisted in evaluating promotional timing and inventory planning for stores. The findings also supported broader omnichannel strategy decisions by revealing customer habits that varied between digital and physical touchpoints.

6. Development of a Customer-Centric Feature Table

A centralized customer feature table was designed and implemented using Snowflake SQL to consolidate all relevant customer-level information into a single structure. The goal was to enable faster and more consistent access to demographic attributes, transaction history, loyalty status, and engagement metrics.

This unified table formed the foundation for downstream analytics, including segmentation, campaign targeting, and churn modeling. Reusable Common Table Expressions (CTEs) were created to extract top brands, categories, and recency metrics at both the Amperity ID and email levels. This work helped simplify complex joins and improved collaboration across teams working with customer data.

7. Standardization of Product Information

Inconsistencies in brand and category naming across product records presented challenges in accurate customer profiling. To resolve this, a standardized mapping between SKUs and product metadata was developed, ensuring uniformity in the identification of top-purchased brands and categories.

This step was crucial in producing clean, consistent features for analytics. It also enabled correct grouping for KPIs like average selling price (ASP), return rate, and units per transaction by category or brand. Clean data contributed to more meaningful insights and eliminated noise in customer preference analyses.

8. Email Subscriber Engagement Analysis

The engagement behavior of email subscribers was analyzed to evaluate the effectiveness of digital outreach campaigns. Metrics such as open rates, click-through rates, and purchase conversions were tracked across time and cohorts. Subscriber activity was correlated with downstream transactions to assess campaign ROI and refine targeting strategies.

This analysis helped identify high-potential segments for re-engagement and confirmed which types of messaging led to stronger customer response. Recommendations were made for adjusting frequency, content themes, and targeting rules in future email campaigns to improve engagement and reduce opt-outs.

9. Identification of Top Brands and Categories per Customer

To support personalization efforts, an analysis was conducted to extract each customer's top three brands and product categories over different time windows (6 months, 12 months, and overall). Frequency of purchases was used as the primary ranking criterion, with recency as a tie-breaker.

This data was integrated into the customer feature table, allowing the marketing team to tailor promotions based on individual preferences. The approach enhanced personalization in both email marketing and website experience, leading to higher customer satisfaction and engagement.

10. Skill Development through Online Courses

In addition to hands-on project work, foundational knowledge was strengthened through the completion of self-paced online courses in key areas such as statistics, Python programming, SQL, and machine learning. These courses enhanced technical skills necessary for real-world applications in data science and marketing analytics.

The theoretical knowledge gained supported better decision-making during tasks like EDA, feature engineering, and predictive modeling. Continuous learning ensured a well-rounded understanding of both the business context and the analytical tools required to succeed in the role.

11. Performance Monitoring through KPI Dashboards

Monitoring and evaluating the effectiveness of marketing and customer engagement strategies required the development of real-time dashboards for key performance indicators (KPIs). Using tools like Python, SQL, and Snowflake, a range of visualizations were built to track metrics such as churn rate, AOV, repeat purchase rate, and lifetime value. These dashboards offered dynamic filtering capabilities, enabling users to segment data by acquisition channel, time period, and customer tier. This made it possible for non-technical teams to self-serve analytics, reducing dependence on ad hoc reporting.

The dashboards not only standardized performance reviews across teams but also encouraged a culture of data-driven decision-making. Business stakeholders could directly interact with the data, compare trends across time windows, and identify areas requiring strategic interventions. This setup also supported scenario analysis and campaign planning, making performance evaluation more actionable and immediate. It empowered leadership with consistent, timely insights, improving alignment between outputs and marketing goals.

12. Loyalty Program Impact Analysis

The loyalty program's contribution to customer retention and profitability was analyzed in depth to determine its long-term effectiveness. Customers were divided into loyalty members and non-members, and behavioral comparisons were made using metrics such as frequency of repeat purchases, time between transactions, average order value, and total lifetime value. It was observed that loyalty members not only transacted more frequently but also showed stronger engagement with marketing campaigns, especially those offering exclusive benefits.

Further cohort analysis revealed that customers who enrolled early in the loyalty program exhibited sustained retention over time. These insights led to strategic recommendations, including the expansion of tiered rewards, early access offers, and personalized loyalty incentives. By quantifying the financial impact of loyal customers, the analysis provided strong justification for further investment in retention through loyalty initiatives, enhancing long-term brand equity and customer satisfaction.

13. Predictive Modeling for Repeat Purchase Likelihood

A predictive model was developed to identify which customers were most likely to make another purchase in the next 90 days. Unlike churn models, this model focused on positive customer behaviors and buying intent. Using classification techniques like logistic regression and decision trees, a probability score was generated for each customer based on features such as last purchase date, email engagement, browsing activity, and spend history. Data preprocessing included normalization, feature engineering, and null-value handling to ensure model reliability.

The model was integrated into dashboards to allow real-time access to "high-opportunity" customers, guiding personalized re-engagement strategies like targeted offers or reminder emails. This use of predictive analytics shifted the business approach from reactive to proactive. The scoring system provided marketing teams with a clearer understanding of customer potential and helped optimize promotional budgets toward segments with higher predicted returns.

14. Integration of Experian Data for Enhanced Profiling

To improve the accuracy of customer segmentation, third-party demographic data from Experian was incorporated into the internal dataset. This enriched the existing profiles with valuable attributes such as income band, marital status, homeownership, and lifestyle segment. Matching this data with internal records required building ID-mapping logic and addressing inconsistencies in naming conventions and formatting. Snowflake SQL was used extensively to join, clean, and validate the appended attributes.

The integration enabled the construction of highly nuanced customer personas, which supported better targeting of luxury products, high-value campaigns, and retention-focused strategies. With deeper insights into customer backgrounds and preferences, marketing messages could be tailored more effectively, increasing campaign resonance. This level of profile enrichment also supported advanced modeling initiatives and offered a strategic edge in competitive segmentation and acquisition.

15. Support in Cross-Functional Collaboration and Knowledge Sharing

This internship offered valuable experience in collaborating across departments, including marketing, product, and data engineering. Regular sync-ups and review meetings helped align data deliverables with evolving business needs. Working in tandem with non-technical stakeholders required simplifying technical outputs into actionable insights, often through storytelling techniques, visual dashboards, and annotated SQL scripts.

The scope of the internship encompassed a comprehensive blend of technical data analytics, strategic marketing support, and cross-functional collaboration. From developing predictive models to building dynamic dashboards and executing loyalty analysis, each component contributed to a stronger understanding of customer behavior and business impact. By working hands-on with advanced tools and real-world datasets, the project enabled the translation of raw data into meaningful insights. This multifaceted experience not only sharpened technical proficiency but also deepened the ability to approach problems from a business-first perspective. Ultimately, the work delivered measurable outcomes and laid the foundation for future data-driven decision-making within the organization.

CHAPTER 4

LEARNING OBJECTIVES

1. To understand and analyse customer retention metrics in a retail environment

One of the primary learning objectives was to develop a strong understanding of customer retention and its importance in the retail industry. This involved studying key metrics such as Average Order Value (AOV), Units per Transaction (UPT), and customer engagement through marketing channels like email. Through hands-on data analysis, insights were gained into how these metrics reflect customer loyalty and help identify the effectiveness of marketing strategies. This objective helped in building a foundational understanding of how businesses measure and manage long-term customer relationships.

2. To apply cohort analysis techniques for segmenting and tracking customer behavior

Another significant goal was to learn and apply cohort analysis techniques to real-world retail data. By grouping customers based on their acquisition period and tracking their behavior over time, patterns in retention, repeat purchases, and drop-off rates could be observed. This method provided a powerful lens to compare different customer groups and understand the impact of time-based marketing strategies. Learning cohort analysis enhanced the ability to interpret customer lifecycle dynamics and evaluate the performance of various acquisition campaigns.

3. To build and evaluate a predictive model for customer churn

A key learning objective was to gain practical experience in building a churn prediction model using machine learning. This included preparing the dataset, selecting relevant features, and applying classification algorithms to predict customer churn. The process involved data preprocessing, model training, validation, and performance evaluation using metrics like accuracy and ROC-AUC. This hands-on work deepened the understanding of how data science techniques can be applied to solve real business problems and highlighted the value of predictive analytics in customer retention strategies.

4. To derive marketing insights and convert them into business recommendations

The final objective was to bridge the gap between data analysis and business strategy by generating insights and translating them into actionable recommendations. This involved interpreting the results of retention analysis, cohort trends, and churn predictions, and presenting them in a format suitable for decision-making. The process included creating reports and visual dashboards to communicate findings effectively. This objective emphasized the importance of storytelling with data and demonstrated how analytics can influence marketing tactics, budgeting, and customer engagement planning.

5. To enhance data visualization and reporting skills for analytical storytelling

An important learning objective was to develop the ability to present complex data insights in a clear and visually engaging manner. This involved using tools such as Excel, Tableau, or Python libraries like Matplotlib and Seaborn to create charts, graphs, and dashboards that highlighted key findings from the retention, cohort, and churn analyses. The goal was not just to analyze the data, but to communicate the results effectively to stakeholders. Through this process, an understanding was gained of how visualization aids in storytelling, supports decision-making, and helps convey actionable insights to both technical and non-technical audiences.

6. To perform exploratory data analysis (EDA) on transactional and store-level datasets

Developing the ability to explore and summarize large volumes of transactional and store-level data was a key objective. EDA techniques were used to uncover purchase trends, channel preferences, and seasonal variations. Statistical analysis and visual tools helped identify customer buying patterns and anomalies in the data. By conducting EDA, a data-driven foundation was created for further analysis such as purchase prediction and segmentation. This objective enhanced the ability to ask the right questions and uncover hidden insights in real-world datasets.

7. To build a consolidated customer-centric data model

Creating a unified customer-level dataset was a critical learning goal. The objective was to integrate disparate data sources—such as transactions, product information, loyalty status, and engagement history—into a single table for analysis. SQL skills were applied to design reusable data pipelines and Common Table Expressions that streamlined feature generation.

This effort improved understanding of the data engineering side of analytics and demonstrated how structured data modeling supports scalable analytics and personalization use cases.

8. To clean and standardize inconsistent product metadata

Handling inconsistencies in product descriptions and category labels was an important technical objective. Through mapping techniques and reference tables, product information was standardized to support accurate analysis of top brands, categories, and customer preferences. This task emphasized the importance of data quality and integrity in downstream analysis. The experience strengthened the ability to clean, preprocess, and structure raw data into analysis-ready formats—an essential step in any data science project.

9. To explore customer purchase prediction using EDA

A key learning was to analyze time-based purchase behavior to estimate when a customer is likely to buy again. Historical transaction data was used to determine the average days between purchases and analyze how channel behavior (store, online, omni) affected likelihood of repurchase. A 90-day repurchase threshold was identified based on data trends. This objective helped develop skills in customer lifecycle analysis and prediction through exploratory techniques before moving into machine learning models.

10. To build domain knowledge in the fashion retail industry through data

A critical learning objective was to gain in-depth domain knowledge of the fashion retail sector by working with actual business data. The internship provided exposure to various retail-specific challenges such as seasonality, shifting consumer preferences, and the dynamics of fast fashion. By analyzing customer transactions, product categories, and engagement across digital and physical touchpoints, valuable insights were gained into how customer behavior changes based on product type, pricing, promotional periods, and purchase channels.

Additionally, the omnichannel nature of the company offered a unique opportunity to study how customer interactions differ across online, in-store, and combined (omni) journeys. This understanding was further deepened by evaluating the performance of loyalty programs, email campaigns, and app-based engagement. Learning how business decisions—such as inventory management, promotional strategies, and personalization—are driven by data added a strong business context to the technical skills being developed.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Data Integration from Multiple Sources

Challenge:

Integrating data from various internal and third-party systems—such as transactional records, customer demographics, Amperity, and Experian—presented major difficulties due to inconsistent formats, mismatched keys, and varying data structures. Each dataset came with different levels of granularity, complicating the process of building a unified view of the customer.

Solution:

A standardized data schema was designed to harmonize formats. Snowflake SQL was used to join and normalize datasets while handling missing or duplicate values with data wrangling techniques. Frequent validation ensured data alignment across platforms, and summary tables were created to simplify downstream processes.

2. Handling Large Volumes of Data

Challenge:

Processing vast amounts of omnichannel data slowed down queries and delayed analytical workflows, especially during exploratory data analysis (EDA) and metric calculations across customer segments.

Solution:

Query optimization, table partitioning, and data aggregation techniques were implemented to improve performance. Summary tables were created to store frequently used metrics. Snowflake's scalable infrastructure helped handle heavy queries efficiently, allowing faster iterations during the analysis.

3. Customer Segmentation Complexity

Challenge:

Segmenting customers meaningfully was difficult due to diverse behaviors across channels and the need to consider multiple variables like frequency, recency, and average basket size.

Solution:

Customer segments were developed using RFM analysis and clustering techniques. Additional segmentation was done based on engagement channels and product preferences. Continuous refinement ensured segments aligned well with business goals and marketing needs.

4. Ensuring Data Quality and Accuracy

Challenge:

Inconsistent data entry, duplication, and missing values posed risks to the accuracy of metrics such as retention rates, AOV, and churn predictions.

Solution:

Automated data cleaning steps were introduced using SQL scripts, and quality checks were applied across datasets. Collaboration with data engineers ensured upstream pipelines delivered cleaner data. Business rules were applied for anomaly handling.

5. Retention Metric Calculation for FY 2024–25

Challenge:

Calculating annual retention metrics such as subscriber activity, app downloads, and loyalty engagement required merging and filtering complex datasets and time-bound events.

Solution:

Date filtering, cohort-based filtering, and event matching logic were applied using SQL to calculate precise retention values. Dashboards were built to visualize these KPIs for monthly and quarterly tracking.

6. Channel Identification for Purchase Behavior

Challenge:

Understanding the impact of online, offline, and omnichannel touchpoints on purchase behavior was challenging due to overlapping customer journeys.

Solution:

Purchase data was categorized by channel using transaction source metadata. EDA was conducted to assess repurchase patterns across channels, identifying that a 90-day window was optimal for defining likely repurchases.

7. Building a Unified Customer-Centric Table

Challenge:

Combining all relevant customer data into a single table for downstream use required managing schema alignment, deduplication, and ensuring consistent customer keys.

Solution:

A customer-centric table was created by joining transactional, demographic, and loyalty data using consistent identifiers. The table was optimized for join operations and validated to support feature extraction and reporting.

8. Extracting Top Brands and Categories per Customer

Challenge:

Identifying the top 3 brands and categories per customer based on frequency and recency required sorting large datasets with varied product metadata.

Solution:

Ranking functions (e.g., ROW_NUMBER, DENSE_RANK) were used along with windowing functions to prioritize purchases. Product IDs were mapped to standardized brand and category fields for consistency, allowing accurate top-N extraction.

9. Dealing with Inconsistent Product Data

Challenge:

Products were mapped inconsistently across different systems, leading to difficulties in aggregating brand and category-level insights.

Solution:

A standardized product-SKU mapping table was used to unify product descriptions. This enabled accurate category-level aggregations and consistent insights generation across datasets.

10. Creating a Scalable Feature Store

Challenge:

As feature requirements evolved, it became difficult to manage different data points—like churn probability, brand affinity, AOV trends—across multiple use cases. Recomputing these metrics for each analysis introduced inefficiencies and inconsistencies.

Solution:

To address this, a reusable and modular feature store was created using Snowflake views and materialized tables. Features were categorized as transactional, demographic, or behavioral, and standardized column naming conventions were followed. SQL templates were developed for new features, and existing ones were updated through version-controlled scripts. This allowed seamless integration with dashboards and models, saving time and ensuring consistency across teams.

11. Managing Null and Missing Data in Customer Records

Challenge:

Missing values in key fields such as age, gender, or loyalty tier posed problems for segmentation, model training, and demographic profiling. Filling them incorrectly could bias results, but excluding records led to smaller sample sizes.

Solution:

Business-driven imputation logic was applied based on known associations. For example, missing gender values were inferred based on first name using reference datasets. Age ranges were imputed using birthdate clusters. Where imputation was risky, flags were added to denote “missingness” as a feature in itself. This dual approach preserved data integrity while enabling full use of the available dataset.

12. Implementing Time-Based Retention Windows

Challenge:

Different departments wanted to analyze retention based on varying timeframes—some used 30-day repurchase, others 60 or 90-day windows. Manually recalculating each metric introduced errors and was time-consuming.

Solution:

Dynamic SQL logic was developed using parameterized windowing functions that allowed defining rolling timeframes based on a reference transaction date. This meant the same script could be reused to calculate retention across any time horizon. Results were stored in a structured format (customer_id, window_type, retention_flag) to support flexible reporting.

13. Validating Business KPIs across Dashboards

Challenge:

Discrepancies were often observed between KPIs reported by different dashboards (e.g., marketing vs. product analytics), causing confusion in meetings and decision-making delays.

Solution:

A master KPI dictionary was created, which included definitions, formulas, and data sources for each metric. This was shared across teams and embedded into documentation for all dashboards. KPI logic was centralized into reusable SQL scripts, and automated checks were set up to compare values across dashboards weekly. This improved confidence in data and reduced redundant debates over metric accuracy.

14. Calculating Unique Customer Contributions per Channel

Challenge:

Attributing value to customers across channels (online, app, offline) was complicated by overlap in purchases and identity resolution issues. Some customers appeared multiple times with different identifiers.

Solution:

Customer consolidation logic was applied using fuzzy matching and deterministic keys (email, phone number, loyalty ID). Contribution was calculated using weighted attribution models: last-click, first-click, and linear attribution models were tested. Insights were layered into dashboards to show how customers transitioned across channels.

15. Handling SKU-Level Transaction Anomalies

Challenge:

Some transactions included SKU-level records with null prices or unrealistic values (e.g., \$0.01 for premium products), likely due to system errors or promotional misclassifications. These records skewed metrics like AOV, UPT, and product performance.

Solution:

A combination of filtering thresholds, business logic, and anomaly detection (z-score analysis on price fields) was applied to flag suspicious records. Promotions were cross-verified with a discount rule engine, and only valid transactions were retained in KPI calculations. A "data quality score" was assigned to each record and reported for future pipeline improvements.

16. Aligning Offline and Online Taxonomies

Challenge:

Product categories and brand names often varied between in-store and online systems. For example, "Footwear" online might be recorded as "Shoes" offline, and some brands were only available on one channel.

Solution:

A unified taxonomy table was created with canonical_category, source_category, and channel columns. Mapping rules were reviewed with domain experts to ensure accuracy. All analysis thereafter referenced this canonical table, ensuring brand and category aggregations were consistent across platforms.

17. Delayed Data Availability from External Vendors

Challenge:

Data from third-party sources like Experian or marketing automation tools often arrived with delays, affecting monthly reports and retention cohort definitions. Late-arriving data also introduced inconsistencies when backfilling historical records.

Solution:

Buffer windows were added to all pipelines, with automated triggers checking data completeness before execution. Stale data flags were introduced in dashboards, and versioned datasets allowed for backdated updates without overriding prior results. Communication protocols were improved with vendors to ensure SLAs were met more reliably.

18. Managing Feature Drift over Time

Challenge:

Features such as average purchase frequency or discount affinity changed as customer behavior evolved. Static features quickly became outdated, leading to model decay and misaligned targeting.

Solution:

Rolling windows (30D, 90D, 6M) were implemented to track temporal changes in key features. Drift detection scripts compared current distributions with historical benchmarks, flagging significant shifts. Features were labeled with timestamps and retrained models were versioned accordingly. This adaptive method ensured features stayed relevant and accurate over time.

19. Collaborating with Cross-Functional Teams for Adoption

Challenge:

Getting buy-in from non-technical teams (like marketing or creative) to use analytical outputs was difficult. They often found dashboards overwhelming or were unsure how to interpret certain metrics.

Solution:

Customized views were created for different teams with simplified KPIs, interactive filters, and explanatory tooltips. Walkthrough sessions were conducted to onboard users. Feedback loops were incorporated through Slack or email, and an FAQ section was maintained for each dashboard. These efforts significantly improved adoption and alignment.

20. Balancing Model Accuracy and Interpretability in Churn Prediction

Challenge:

While building churn prediction models, complex algorithms like XGBoost or Random Forests provided high accuracy but were difficult for stakeholders to interpret. Non-technical teams wanted to understand why a customer was being flagged as high risk, but black-box models didn't offer clear explanations. This created a communication gap between data science and business teams, slowing adoption and trust in model outputs.

Solution:

A two-pronged approach was adopted. First, a simplified logistic regression model was developed in parallel with the complex models to serve as a baseline explanation tool. This interpretable model helped highlight key factors like last purchase date, discount usage, and frequency drop. Second, SHAP (SHapley Additive exPlanations) values were used with the complex models to explain individual predictions in a human-readable format. These SHAP summaries were incorporated into dashboards, enabling teams to see why each customer was flagged, and what actions could be taken. This balance ensured that models remained both powerful and transparent, improving trust and enabling targeted retention efforts.

The challenges encountered during the internship offered invaluable lessons in managing real-world data complexities and building scalable analytics solutions. From integrating disparate datasets and maintaining data quality to developing interpretable models and extracting actionable insights, each hurdle strengthened both technical proficiency and strategic thinking. Tackling these problems not only enhanced problem-solving and critical analysis skills but also emphasized the importance of communication and collaboration across teams. The experience demonstrated that effective data analytics goes beyond code—it requires a deep understanding of business needs, a commitment to data accuracy, and the ability to translate technical outputs into meaningful business strategies. Overall, these challenges served as essential stepping stones in the journey of becoming a well-rounded data professional.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Successful Implementation of Cohort-Based Retention Analysis

One of the key achievements during the internship was the successful application of cohort analysis to segment customers based on acquisition time and observe their behavioral trends over months. The analysis provided critical insights into retention drop-off points and highlighted the effectiveness of previous marketing strategies. This contribution enabled the business to make informed decisions regarding customer lifecycle management and design data-driven engagement campaigns.

2. Development of a Churn Prediction Model

A significant contribution was the design and development of a machine learning model to predict customer churn. The model was trained using relevant customer behavior features such as transaction frequency, average order value, and recency of last purchase. The output provided a churn probability score, allowing the marketing team to identify high-risk customers and take proactive steps to re-engage them. This model is a valuable asset for customer retention efforts.

3. Data Cleaning and Feature Engineering for Business Readiness

An important aspect of the project involved extensive data preprocessing and feature engineering to prepare the dataset for meaningful analysis and modeling. This included handling missing values, removing duplicates, and creating new derived features like customer tenure and average basket size. The cleaned and transformed dataset not only improved model accuracy but also ensured that the insights generated were business-ready and actionable.

4. Insightful Marketing Reports and Visual Dashboards

Another major contribution was the creation of marketing insight reports and interactive dashboards that presented analytical findings in a visually compelling manner. These reports included key metrics on customer retention, cohort performance, and churn predictions, making it easier for marketing teams to track performance and identify areas for improvement. The visualization work bridged the gap between data science and business application.

5. Practical Exposure to Real-World Retail Data and Strategy

The internship provided hands-on experience with real-world data from the retail sector, allowing for an in-depth understanding of customer behavior, sales trends, and marketing effectiveness. Contributing to strategic decision-making through data-backed insights gave a real sense of how analytics directly supports business goals. This practical exposure helped build both technical confidence and business acumen.

6. Enhanced Understanding of Customer Lifecycle Management

Through continuous analysis of customer behavior over various stages — acquisition, engagement, retention, and churn — a deeper understanding of the customer lifecycle in the retail industry was developed. This learning was translated into actionable insights that contributed to designing more targeted and timely marketing strategies. By aligning the business goals with customer journey data, the contribution strengthened the customer-centric approach of the organization.

7. Collaboration with Cross-Functional Teams for Insight Alignment

Effective collaboration with marketing, data, and business intelligence teams played a vital role in aligning analytical findings with strategic objectives. Contributions included participating in discussions, understanding business needs, and adjusting analysis techniques based on team feedback. This cross-functional collaboration ensured that the analytics work was relevant, timely, and impactful, further enhancing the overall value delivered to the organization.

8. Creation of a Customer-Centric Master Table

A major technical contribution was the development of a consolidated customer-centric table that unified demographic, behavioral, and transactional data at the customer level. This table served as a single source of truth for downstream analysis and reporting. It streamlined access to key customer metrics and enhanced the efficiency of ongoing marketing and analytics operations.

9. Identification of Purchase Likelihood Window

By conducting exploratory data analysis (EDA) on Omni channel transactional data, an optimal re-purchase window of 90 days was identified. This insight allowed the business to better time re-engagement efforts and personalize follow-up strategies. It contributed to more precise marketing campaigns aimed at increasing repeat purchases and customer retention.

10. Strengthening Analytical Skills through Online Learning

To support project goals, relevant online courses in statistics, Python, SQL, and machine learning were completed. This proactive learning enriched the technical toolkit necessary for data retrieval, cleaning, EDA, and modeling tasks. The enhanced skills directly contributed to the success and accuracy of the analytics work delivered during the internship.

11. Standardization of Brand and Category Hierarchies

Contributed to the consistency of analytical outputs by mapping inconsistent brand and category names through SKU-to-product ID standardization. This ensured accurate aggregation of product-level metrics and supported the generation of top brands/categories per customer segment. The effort improved the reliability of insights used in segmentation and personalization.

12. Automation of Data Pipelines for Efficiency

Helped design repeatable SQL structures and reusable CTEs to automate parts of the data transformation and feature creation process. This not only reduced manual work but also ensured consistency in metric definitions across multiple reports and analyses. The automation streamlined the analytics workflow and improved turnaround times.

13. Support for Email-Level Insights across Amperity IDs

Extended feature engineering logic to aggregate brand preferences at the email level, combining insights from multiple Amperity IDs. This contribution helped unify customer profiles across identities and enhanced personalization efforts at the marketing level. The output enabled more informed cross-device and cross-platform engagement strategies.

14. Optimization of Query Performance for Large Datasets

Worked on optimizing Snowflake SQL queries for faster execution while dealing with large-scale transactional and customer datasets. Techniques like filtering early, using efficient joins, and working with summary tables led to improved query performance. This technical refinement increased the speed and reliability of analyses.

15. Application of Data-Driven Thinking to Marketing Strategy

Actively contributed to the transformation of raw data into meaningful business strategy by linking customer behavior trends with campaign decisions. The application of data-driven thinking helped prioritize high-retention cohorts and supported budget reallocation.

16. Optimization of SQL Queries for Improved Performance

During the course of data extraction and transformation, several complex SQL queries were written to handle large datasets. Through iterative testing and optimization techniques—such as using CTEs, appropriate joins, and indexed columns—the performance of these queries was significantly improved. This reduced execution time, improved pipeline efficiency, and enabled faster insight generation for business users.

17. Standardization of Brand and Category Mapping

A major contribution was the creation of a standardized mapping process to resolve inconsistencies in brand and category names sourced from item-level transaction data. By linking product IDs with a master SKU-to-brand/category table, consistent labeling was ensured across all analyses. This directly improved the accuracy of top-N product analytics and reporting used by the merchandising and marketing teams.

18. Creation of a Reusable Feature Store for Retention Modeling

Contributed to the development of a reusable feature store in Snowflake by engineering key features such as customer recency, frequency, and product preferences over different time windows (e.g., 6M, 12M, lifetime). This structure allowed easy plug-and-play access for downstream modeling and experimentation, saving time and ensuring consistency across projects.

19. Integration of Third-Party Data for Enriched Customer Profiles

Assisted in integrating data from third-party platforms like Amperity and Experian to enrich internal customer profiles with demographic, behavioral, and engagement attributes. This enhancement added depth to customer understanding and enabled more precise segmentation, targeting, and lookalike modeling.

20. Automation of Retention Metric Dashboards

Led the automation of retention-related dashboards using Tableau and Snowflake views, reducing manual refresh efforts and enabling near real-time tracking of KPIs such as repeat purchase rate, cohort survival rate, and lifetime value. This automation helped business users stay updated and make timely decisions based on current data trends.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship provided a transformative opportunity to apply academic knowledge to a real-world business setting. Working with customer and transactional data in the retail domain offered valuable exposure to the complexities of consumer behavior. Concepts like cohort analysis, retention metrics, and churn modeling were explored hands-on, deepening both technical and analytical understanding. The experience served as a strong introduction to data-driven decision-making in a fast-paced industry.

An important realization during the internship was the critical role of data quality. While theoretical models often assume clean datasets, real-world data comes with imperfections missing values, inconsistent formats, and duplicates. Time invested in preprocessing, cleaning, and validating data proved to be essential for reliable outputs. This process taught the importance of patience, detail orientation, and structured thinking when dealing with data at scale.

Another area of growth was feature engineering and metric design. Creating variables such as customer tenure, frequency bands, and brand preference rankings allowed the raw data to be transformed into insights. Learning to identify meaningful features helped in understanding which aspects of customer behavior truly influence retention and engagement. This skill is foundational for any analytics role and was significantly sharpened through practical exposure.

Technical proficiency improved notably during the internship. Tools such as Snowflake SQL, Python (for modeling and visualization), and Excel (for quick data views) became central to the workflow. Visualization libraries like Seaborn and Matplotlib were used to present trends effectively, while SQL queries were optimized for better performance. These tools enabled both in-depth analysis and fast iteration, increasing overall efficiency and output quality.

Beyond technical skills, the internship enhanced business acumen and strategic thinking. Every analysis conducted had a purpose—whether to reduce churn, improve engagement, or personalize customer outreach. Understanding the business goals behind the data tasks fostered a more structured and purposeful approach to analytics. It emphasized the need to align data work with broader organizational outcomes.

Working on the churn prediction model helped deepen my knowledge of machine learning and data science techniques. It was my first time developing a predictive model from start to finish in a business context. I was involved in selecting features such as purchase frequency, recency, and average order value, as well as experimenting with classification algorithms to identify which customers were at risk of disengaging. I also learned to evaluate the model using metrics such as accuracy, precision, and ROC-AUC scores.

Beyond just building the model, I understood how interpretability matters—identifying what features contribute most to churn helps the business create more effective interventions. One of the highlights was collaborating with cross-functional teams, including marketing and data engineering. These interactions highlighted how data projects are rarely executed in isolation—they require communication, alignment, and adaptability. Feedback from stakeholders led to refining models and adjusting queries, teaching the value of iteration and flexibility in a business environment.

Creating reports and dashboards for non-technical users revealed the importance of communication. Turning complex analysis into visually intuitive formats helped bridge the gap between data and decision-making. This experience improved storytelling skills and demonstrated how the real impact of analytics lies not just in accuracy, but in clarity and usability.

Ultimately, the internship reinforced a deep interest in marketing analytics and customer behavior modeling. The exposure to real-world retail data and business operations created a strong foundation for future career growth. It instilled confidence, encouraged continuous learning, and built the discipline needed to solve analytical challenges with both rigor and creativity.

CHAPTER 8

RECOMMENDATIONS

1. Invest in Personalized Marketing Campaigns

To improve customer retention, the company should consistently invest in personalized marketing strategies. By leveraging behavioral, transactional, and demographic data, marketing messages can be tailored to resonate with individual customer preferences. Personalized promotions, curated product suggestions, and targeted email campaigns foster a stronger emotional connection with the brand and lead to increased repeat purchases.

2. Enhance Customer Segmentation Techniques

Segmentation based on RFM (Recency, Frequency, and Monetary) and engagement patterns enables more precise targeting. Creating micro-segments for different types of customers—such as loyal, dormant, or new—can guide the company in delivering more relevant and timely messages. This approach ensures marketing efforts are not only efficient but also customer-centric, increasing retention and satisfaction.

3. Implement Real-Time Churn Monitoring

The company should consider integrating real-time churn prediction models into its CRM systems. These models can continuously score customers based on their risk of attrition, enabling the business to trigger timely interventions. Proactive measures such as re-engagement emails, loyalty rewards, or personalized outreach can help retain high-risk customers and reduce overall churn.

4. Automate Cohort Analysis in Reporting Pipelines

Standardizing and automating cohort analysis as a part of regular analytics reporting will provide consistent insights into customer behavior over time. This allows teams to track how different acquisition cohorts respond to marketing efforts and identify long-term trends in retention and spending. Automation reduces manual effort and ensures timely access to actionable insights.

5. Adopt a Centralized Data Infrastructure

As data grows in volume and complexity, it is essential to maintain a clean and scalable data infrastructure. Investing in a centralized data warehouse enables efficient data integration across multiple platforms and departments. A unified data view not only supports accurate analysis but also enhances collaboration across teams.

6. Enable Self-Service Analytics through Dashboards

Building user-friendly, interactive dashboards for key performance indicators such as churn rate, AOV, and customer retention can empower non-technical teams. With access to real-time insights, teams can independently monitor performance and make informed decisions. Tools like Tableau or Power BI should be used to support this initiative.

7. Improve Cross-Functional Data Literacy

Training programs should be introduced to upskill teams in basic data literacy and analytical thinking. When marketing, sales, and product teams are data-aware, they can better interpret analytics reports and contribute to strategy discussions. A shared understanding of data fosters alignment and speeds up decision-making across functions.

8. Utilize A/B Testing for Campaign Optimization

To enhance the effectiveness of retention campaigns, the company should implement structured A/B testing. By comparing variations of promotional content, timing, and messaging, teams can learn what resonates best with different segments. These insights will refine future marketing strategies and lead to more impactful customer interactions.

9. Track Customer Lifecycle Stages Continuously

It is recommended to monitor and map customers through various stages of the lifecycle-acquisition, engagement, growth, and reactivation. Understanding transitions between stages enables timely interventions and targeted strategies that address customers' evolving needs. Lifecycle tracking also helps align retention efforts with business objectives.

10. Establish Feedback Loops Between Analytics and Strategy Teams

Finally, creating continuous feedback mechanisms between analytics teams and decision-makers ensures that data insights are applied effectively. Regular meetings, collaborative dashboards, and shared success metrics will help align analytics work with real-time business needs, resulting in more agile and impactful strategies.

11. Incorporate Predictive Lifetime Value (LTV) Modeling

A critical recommendation is the adoption of predictive Lifetime Value (LTV) models to forecast the long-term revenue potential of each customer. By analyzing historical transaction data, engagement frequency, recency, and product preferences, businesses can classify customers into high, medium, and low-value brackets.

This segmentation helps marketing and retention teams prioritize their efforts and budget effectively. High LTV customers, for example, can be offered premium services, early access to sales, or personalized thank-you campaigns to deepen loyalty. On the other hand, customers with declining value can be re-engaged with customized incentives. Integrating LTV scores into CRM systems ensures every customer interaction is strategically aligned with their future value, allowing the organization to focus on sustainable growth and profitability.

12.Leverage Omni channel Data for Unified Customer Profiles

To enable seamless and relevant customer experiences, it is recommended that the organization create unified customer profiles by integrating data from all touchpoints—including in-store visits, online shopping, mobile app usage, social media engagement, and customer service interactions. Often, businesses treat these channels in silos, leading to fragmented communication and inconsistent personalization. A centralized customer data platform (CDP) can consolidate Omni channel data, providing a 360-degree view of each customer's behavior, preferences, and history. With this comprehensive view, marketing strategies can become more intelligent—offering tailored recommendations across every platform and ensuring that the brand voice remains consistent. This approach not only improves personalization but also reduces redundancy and enhances the overall customer experience, fostering stronger relationships and loyalty.

13.Launch Loyalty and Reward Programs Based on Data Insights

Loyalty programs, when crafted using data-driven insights, can become powerful tools for customer retention. The company should leverage behavioral and transactional data to design dynamic loyalty schemes that reward customers based on their actual engagement levels, purchase frequency, and historical value. Instead of offering uniform benefits, customers can be categorized into tiers where higher engagement unlocks exclusive perks such as free shipping, birthday discounts, early access to new collections, or personalized gift cards. Gamifying loyalty—through point systems or milestone achievements—can also boost participation. Furthermore, monitoring the performance of loyalty programs using cohort analysis ensures that they are continuously optimized for maximum effectiveness. A smart loyalty program encourages repeat purchases and turns satisfied buyers into brand advocates.

14.Implement Trigger-Based Communication Workflows

The organization can significantly enhance its marketing effectiveness by establishing automated, trigger-based communication flows. These workflows respond to customer actions or inactions in real-time—such as cart abandonment, repeat purchases, product views, or prolonged inactivity—by sending relevant and timely messages. For instance, a customer who leaves items in their cart can receive a gentle reminder or an incentive offer, while a frequent buyer can be greeted with appreciation emails or loyalty bonuses. By using tools like marketing automation platforms integrated with behavioral data, businesses can deliver contextual communications at scale. These real-time nudges help maintain engagement, recover lost revenue opportunities, and reinforce positive brand associations. Over time, such automation reduces manual workload while ensuring consistent and intelligent customer outreach.

15.Optimize Mobile Engagement Strategies

In today's mobile-first world, it's essential for businesses to tailor their retention and engagement strategies specifically for mobile users. Customers interact with brands through apps, mobile websites, and SMS on a daily basis, and failing to optimize these touchpoints can lead to missed opportunities. The company should invest in responsive design, mobile push notifications, and in-app personalization to capture attention in short attention spans. For example, sending real-time order updates, exclusive app-only discounts, or product reminders can significantly boost engagement rates. Additionally, analyzing app behavior—such as time spent, pages viewed, and clicks—provides actionable insights for improving UI/UX. When combined with personalized offers, a well-optimized mobile experience not only drives conversions but also contributes to long-term customer retention by creating a seamless and convenient shopping journey.

16.Monitor and Reduce Friction Points in the Customer Journey

Customers today expect seamless and intuitive experiences across all brand interactions. Any point of friction—such as confusing navigation, slow checkout processes, lack of payment options, or delayed customer support—can lead to dissatisfaction and churn. The organization should proactively monitor the end-to-end customer journey using tools like journey mapping, session recordings, and feedback loops. By identifying moments where customers drop off or express frustration, businesses can implement targeted improvements. For instance, simplifying the returns process or offering live chat support during checkout can remove major barriers. These optimizations, while sometimes small, have a cumulative effect in enhancing the overall brand experience.

17. Conduct Regular Customer Satisfaction Surveys

Gathering direct customer feedback is essential for any business aiming to improve retention. Implementing regular surveys, such as Net Promoter Score (NPS), Customer Satisfaction (CSAT), and post-interaction feedback, provides real-time insights into what customer's value and where they feel underserved. These surveys can be distributed through multiple channels—email, SMS, or in-app prompts—ensuring wide reach and inclusivity. Analyzing survey results helps detect patterns, such as declining satisfaction in a particular product category or delivery issue complaints. Taking swift action based on feedback, and communicating the improvements to customers, strengthens trust and shows responsiveness. Beyond quantitative scores, open-ended responses offer rich qualitative data that can inspire innovation and better personalization, keeping customers engaged and loyal.

18. Develop Custom Metrics for Deeper Insights

While standard KPIs like churn rate, AOV, and repeat purchase rate are useful, developing custom metrics tailored to the business's unique goals can offer deeper insights. Metrics like "time to second purchase," "engagement-to-conversion ratio," or "discount sensitivity index" can reveal underlying behaviors not captured by generic KPIs. For example, tracking the speed at which new users make their second purchase can inform onboarding strategies, while monitoring the use of promo codes can help identify price-conscious segments. These advanced metrics should be integrated into dashboards and reviewed regularly with relevant stakeholders. They allow the business to respond to nuanced trends and design highly targeted retention campaigns that reflect customer behaviors more accurately, leading to smarter and more profitable decisions.

19. Collaborate with Customer Support for Retention Insights

Customer support teams interact with buyers at critical moments—when issues arise, when expectations are unmet, or when customers need assistance. These interactions are often rich with data that can inform retention strategies. The organization should establish close collaboration between analytics and support teams to regularly review ticket trends, complaint categories, and customer sentiment. Patterns such as repeated issues with a product line or common complaints about delivery delays should be shared with the relevant departments for immediate action. Additionally, tracking resolution time and satisfaction ratings can help identify friction areas in the service process. When businesses demonstrate responsiveness and care, even during problems, it enhances customer trust and encourages loyalty turning potential detractors into promoters.

20. Benchmark Retention Performance against Industry Standards

Finally, the company should regularly compare its customer retention metrics against industry peers and benchmarks. This competitive analysis highlights where the business is excelling and where improvement is needed. Resources such as annual industry reports, third-party data providers, and customer analytics platforms often provide benchmark values for metrics like customer lifetime value, retention by channel, and churn rate. By identifying areas of underperformance, the company can prioritize strategic investments in those aspects—whether it's mobile UX, personalization, or delivery speed. Benchmarking also supports goal-setting, encourages internal accountability, and fosters a culture of continuous improvement. When businesses stay informed about the competitive landscape, they're better equipped to innovate and deliver standout customer experiences that retain and delight.

The recommendations outlined provide a comprehensive roadmap for enhancing customer retention and deepening engagement through data-driven strategies. By leveraging predictive analytics, optimizing omni channel experiences, and fostering a culture of continuous feedback and collaboration, the organization can create a more personalized, efficient, and customer-centric approach. Implementing these initiatives not only addresses immediate retention challenges but also builds long-term loyalty, increases customer lifetime value, and supports sustainable business growth in an increasingly competitive retail landscape. A commitment to data excellence, innovation, and customer empathy will be key to turning these recommendations into measurable success.

CHAPTER 9

CONCLUSION

The internship experience has provided a comprehensive understanding of how data-driven decision-making plays a vital role in shaping marketing strategies within the retail industry. By working on real-world customer data, key areas such as customer retention, cohort-based segmentation, and churn prediction were explored in depth. These analytical approaches revealed valuable insights into consumer behavior and business performance, demonstrating the power of marketing analytics in driving strategic outcomes.

Through hands-on involvement in data preprocessing, feature engineering, model development, and visualization, technical competencies were significantly enhanced. At the same time, the ability to interpret and present data in a business-relevant manner contributed to the development of effective communication skills. These dual capabilities—technical expertise and business understanding—proved essential in transforming raw data into meaningful and actionable insights.

The project also presented a range of challenges, from data quality issues to handling class imbalance in churn models. Overcoming these hurdles fostered a more solution-oriented mindset and instilled confidence in dealing with real-world analytical tasks. The guidance received from mentors and collaboration with team members played a crucial role in refining both technical output and strategic thinking.

Overall, the internship has been an invaluable step toward building a career in analytics. It offered not only a platform to apply classroom knowledge in a practical setting but also a deeper appreciation of how businesses leverage data to enhance customer experience, optimize operations, and achieve long-term growth. The learnings, achievements, and insights gained during this journey will serve as a strong foundation for future professional endeavors.

REFERENCES

- [1] <https://www.saksoff5th.com/>
- [2] <https://www.saksoff5th.com/s/SaksOff5th/aboutus.html?sre=footer-saksfirst>
- [3] Fader, P. S., & Hardie, B. G. S. (2009). Customer-Base Valuation in a Contractual Setting: The Perils of Ignoring Heterogeneity. *Marketing Science*, 28(1), 85–98.
<https://doi.org/10.1287/mksc.1080.0395>
- [4] Kotler, P., & Keller, K. L. (2016). *Marketing Management* (15th Ed.). Pearson Education.
- [5] Snowflake Inc. (2024). Snowflake Documentation: Best Practices for Data Warehousing and Query Performance. <https://docs.snowflake.com>
- [6] Amperity. (2024). Customer 360 and Segmentation Guide.
<https://amperity.com/resources>